



Interview Cheat Sheet

Frontend

Easy

1. What is the difference between `==` and `===` in JavaScript?
 2. Explain the difference between `null` and `undefined`.
 3. What are semantic HTML tags and why are they important?
 4. Difference between inline, inline-block, and block elements.
 5. What is the difference between `var`, `let`, and `const`?
 6. Explain the CSS Box Model.
 7. What is the difference between relative, absolute, fixed, and sticky positioning in CSS?
 8. Difference between synchronous and asynchronous JavaScript.
 9. What is the difference between `localStorage`, `sessionStorage`, and cookies?
 10. What is event bubbling and event capturing in JavaScript?
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Medium

1. Explain React's virtual DOM and how it works.
 2. What are React hooks? Give examples.
 3. Difference between `useEffect` and `useLayoutEffect`.
 4. How does React handle reconciliation?
 5. Explain the difference between controlled and uncontrolled components.
 6. What is a service worker, and how does it help in PWA?
 7. How does the browser rendering pipeline work (reflow, repaint, composite)?
 8. Explain CORS and how to handle it on frontend.
 9. Difference between SSR, CSR, and SSG.
 10. What are Web Components and Shadow DOM?
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Senior

1. How does React Fiber improve rendering compared to old reconciliation?
 2. Explain hydration in React SSR.
 3. How do you optimize large lists in React?
 4. What is code splitting and how do you implement it?
 5. Explain how browsers optimize JavaScript execution (JIT, parsing).
 6. How does tree shaking work in bundlers?
 7. How would you reduce bundle size in a React app?
 8. What are the drawbacks of client-side rendering at scale?
 9. How do you implement micro-frontend architecture?
 10. What are advanced patterns for managing state in React (beyond Redux/Context)?
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Rare but Asked

1. Difference between repaint and reflow.

2. How does DNS resolution affect frontend performance?
 3. What is the difference between `async` and `defer` in script tags?
 4. How does browser caching work (ETag, Cache-Control, etc.)?
 5. What's the difference between HTTP/1.1, HTTP/2, and HTTP/3 for frontend?
 6. Explain CSS specificity rules in depth.
 7. How do you prevent layout shift (CLS) in web apps?
 8. Explain `requestAnimationFrame`.
 9. What are WebSockets and how do they differ from HTTP?
 10. How does a browser handle painting text on screen?
 11. What are critical rendering paths?
 12. Difference between inline styles, CSS-in-JS, and external CSS.
 13. How do browsers optimize painting when multiple CSS animations are applied?
 14. How does React's concurrent mode work?
 15. What is RAIL performance model?
 16. Explain prefetch, preload, and preconnect.
 17. How does lazy loading of images/scripts work?
 18. Why do some websites use base64-encoded images?
 19. What is the difference between hydration mismatch and reconciliation issue?
 20. How does Brotli compression improve frontend performance?
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Backend

Easy

1. What is REST API?
2. Difference between GET, POST, PUT, PATCH, and DELETE.

3. What is middleware in Express.js?
 4. Difference between SQL and NoSQL databases.
 5. What is an ORM? Give examples.
 6. Explain the concept of request and response cycle in Node.js.
 7. What is the difference between synchronous and asynchronous functions in Node.js?
 8. What is JWT and why is it used?
 9. Difference between authentication and authorization.
 10. What is load balancing?
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Medium

1. Explain event loop in Node.js.
 2. What is clustering in Node.js and why is it needed?
 3. Difference between horizontal and vertical scaling.
 4. Explain ACID properties in databases.
 5. What is database indexing and how does it work?
 6. Explain CAP theorem.
 7. How do transactions work in SQL databases?
 8. What is rate limiting and how would you implement it?
 9. Difference between Redis and traditional databases.
 10. What are message queues (e.g., RabbitMQ, Kafka) and why are they used?
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Senior

1. Explain how Node.js handles concurrency under the hood.
2. How do you design a highly available microservices architecture?
3. How does database sharding work?
4. Explain distributed transactions and 2-phase commit.

5. How does replication work in MongoDB and PostgreSQL?
 6. Explain gRPC vs REST vs GraphQL.
 7. How do you secure APIs at scale?
 8. What are eventual consistency and strong consistency in distributed systems?
 9. How do you debug memory leaks in Node.js?
 10. How do you implement observability (logs, metrics, tracing) in backend systems?
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Rare but Asked

1. How does DNS affect backend services?
2. Explain how HTTPS handshake works.
3. What is HMAC and how is it used?
4. Difference between symmetric and asymmetric encryption.
5. How does OAuth2 flow work (Authorization Code flow)?
6. How does TLS termination work in load balancers?
7. Difference between optimistic and pessimistic locking.
8. Explain Zookeeper and its role in distributed systems.
9. How does consistent hashing work?
10. Difference between monolith, SOA, and microservices.
11. What is circuit breaker pattern in microservices?
12. How does Raft consensus algorithm work?
13. Difference between Docker container and VM.
14. How does Kubernetes handle scaling?
15. Explain leader election in distributed systems.
16. What is eventual consistency in DynamoDB?
17. How does Kafka guarantee message ordering?

18. Difference between synchronous and asynchronous replication.
19. What are sidecar containers?
20. How do you debug high-latency issues in backend systems?